1. Legendre's Symbol.

linear congruence $ax \equiv b \pmod{n}$

quadratic congruence $x^2 \equiv a \pmod{n}$

$$ax^2 + bx + c \equiv 0 \pmod{n} \quad \times 4a$$

$$4a^2x^2 + 4abx + 4ac \equiv 0 \pmod{4an}$$

$$(2ax+b)^2 \equiv b^2 - 4ac \pmod{4an}$$

$$y^2 \equiv d \pmod{4an}$$

$a \in \mathbb{Z}$, $n \in \mathbb{N}$ $(a, n) = 1$

a is $\begin{cases} \text{quad residue} : x^2 \equiv a \pmod{n} \text{ solvable.} \\ \text{quad non-residue} : \text{otherwise} \end{cases}$

$(a, p) = 1$

Legendre symbol $\left(\frac{a}{p}\right) = \begin{cases} +1 : \text{quad residue} \pmod{p} \\ -1 : \text{quad non-residue} \pmod{p} \end{cases}$

p prime

Note. $a \equiv a' \pmod{p}$ then $\left(\frac{a}{p}\right) = \left(\frac{a'}{p}\right)$

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2. Euler Criterion

odd prime p with $r = \frac{1}{2}(p-1)$ $\Rightarrow$ $\left(\frac{a}{p}\right) \equiv a^r \pmod{p}$

(Fermat) $a^r \equiv \pm 1 \pmod{p}$

(Lagrange) $a^r \equiv 1 \pmod{p}$ has at most r solutions

$2r+1$ reduced set of residues $= \{1, \dots, r, (-r), \dots, (-1)\}$

$\Rightarrow$ $r+1$ quad residues $= \{1^2, \dots, r^2\} = \{g^2, \dots, g^{2r}\}$ with primitive root g

$1^2 = (-1)^2$, $r^2 = (-r)^2$

$\Rightarrow$ $r+1$ quad non-residues

quad residues $\Rightarrow$ $x^2 \equiv a \pmod{p}$

Fermat $\Rightarrow$ $x^{2r} \equiv \underbrace{a^r \equiv 1}_{\text{Euler Criterion}} \pmod{p}$

quad non-residues & Lagrange $\Rightarrow$ $\underbrace{a^r \equiv -1}_{\text{Euler Criterion}} \pmod{p}$

$\boxtimes$

① $\left(\frac{a}{p}\right)\left(\frac{b}{p}\right) = \left(\frac{ab}{p}\right)$

② $\left(\frac{-1}{p}\right) = (-1)^{\frac{1}{2}(p-1)}$

$$= \begin{cases} +1 & \text{if } p \equiv 1 \pmod{4} \\ -1 & \text{if } p \equiv 3 \pmod{4} \end{cases}$$

i.e. $x = \pm(r!)$ (Wilson)

3. Gauss Lemma.

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numerically least residue of $a \pmod{n}$

① $a' \equiv a \pmod{n}$

② $-\frac{1}{2}n < a' \le \frac{1}{2}n \pmod{n}$

(Gauss) $\left(\frac{a}{p}\right) = (-1)^{l}$ where l is # of $a_j < 0$

$(a, p) = 1$, odd prime p

a_j : numerically least residue of ja $(1 \le j \le r)$, $r = \frac{1}{2}(p-1)$

$|a_j|$ $(1 \le j \le r)$ equals $1, 2, \ldots, r$ in some order.

① $1 \le |a_j| \le r$

② distinct $|a_j|$ because $|a_j| \equiv |a_k|$ means $|j|\cancel{a} \equiv |k|\cancel{a} \pmod{p}$

$$(-1)^{l} \cdot (1 \cdots r) = (a_1 \cdots a_r)$$
$$\equiv (1 \cdot a) \cdots (r \cdot a)$$
$$\equiv (r!)(a^r) \pmod{p}$$

therefore $\underbrace{(a^r)}_{\text{Euler Criterion}} \equiv (-1)^{l} \pmod{p}$

$$\left(\frac{2}{p}\right) = (-1)^{\frac{1}{8}(p^2-1)} = \begin{cases} +1 & \text{if } p \equiv \pm 1 \pmod 8 \\ -1 & \text{if } p \equiv \pm 3 \pmod 8 \end{cases}$$

$$\because \quad a_j = \begin{cases} 2j & 1 \le j \le \frac{1}{2}p \\ 2j - p & \frac{1}{2}p < j \le r \end{cases} \xrightarrow{\text{count}} l \equiv \frac{1}{8}(p^2-1) \pmod 2$$

- $S = \frac{1}{8}(p^2-1) = (1 + 2 + \cdots + r)$
- $2S \equiv (2\cdot 1 + 2\cdot 2 + \cdots + 2r) \equiv \underbrace{(1, \ldots, r}_{\text{홀에서 짝수}}) + \underbrace{(-1, \ldots, -r}_{\text{홀에서 홀수}})$

$$\equiv \underbrace{(1, \ldots, r}_{\text{홀에서 짝수}}) + l \cdot p - \underbrace{(1, \ldots, r}_{\text{홀에서 홀수}})$$

therefore $\frac{1}{8}(p^2-1) \equiv l \pmod{2}$ (modulo 2)

4. Law of quad reciprocity

p, q distinct odd primes $\Rightarrow \left(\frac{p}{q}\right)\left(\frac{q}{p}\right) = (-1)^{\frac{1}{4}(p-1)(q-1)}$

with $\left(\frac{p}{q}\right) = \left(\frac{q}{p}\right)$ if $p \equiv 1$ or $q \equiv 1 \pmod 4$

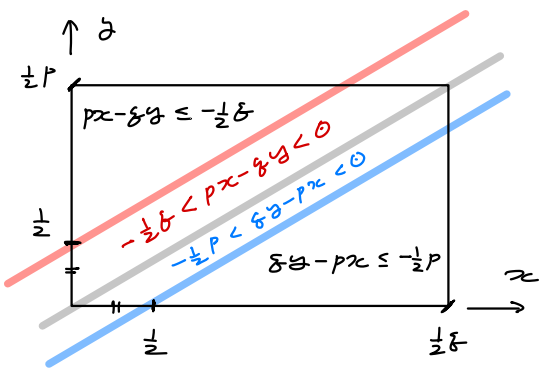
$\left(\frac{p}{q}\right) = -\left(\frac{q}{p}\right)$ if $p \equiv 3$ and $q \equiv 3 \pmod 4$

$\left(\frac{p}{q}\right) = (-1)^{l}$ $\quad l$ is # of lattice points (x, y)

by Gauss

s.t $\begin{cases} 0 < x < \frac{1}{2}q & \text{i.e } j \text{ 제약조건} \\ -\frac{1}{2}q < px - qy < 0 & \text{i.e } aj \text{ 제약조건} \end{cases}$

with $y < \frac{px}{q} + \frac{1}{2} < \frac{1}{2}(p+1)$



$l + m \equiv \frac{1}{4}(p-1)(q-1) - (\triangledown + \triangle) \pmod 2$

bijection

$\begin{cases} x = \frac{1}{2}(q+1) - x' \\ y = \frac{1}{2}(p+1) - y' \end{cases}$

$\left(\frac{15}{71}\right) = \left(\frac{3}{71}\right)\left(\frac{5}{71}\right) = -\left(\frac{71}{3}\right)\left(\frac{71}{5}\right) = -\left(\frac{2}{3}\right)\left(\frac{1}{5}\right) = 1$

$\left(\frac{-3}{p}\right) = \left(\frac{-1}{p}\right)\left(\frac{3}{p}\right) = (-1)^{\frac{1}{2}(p-1)}\left(\frac{3}{p}\right) = \left(\frac{p}{3}\right) = \begin{cases} 1 & \text{if } p \equiv 1 \pmod 6 \\ -1 & \text{if } p \equiv -1 \pmod 6 \end{cases}$

5. Jacobi Symbol

positive odd integer $n = p_1 \cdots p_k$ (products of primes : not necessarily distinct)

Jacobi Symbol

$(a,n) = 1$ then $\left(\frac{a}{n}\right) = \left(\frac{a}{p_1}\right) \cdots \left(\frac{a}{p_k}\right)$

$n = 1$ then $\left(\frac{a}{n}\right) = 1$

$(a,n) > 1$ then $\left(\frac{a}{n}\right) = 0$

Note. $a \equiv a' \pmod{n}$ then $\left(\frac{a}{n}\right) = \left(\frac{a'}{n}\right)$

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a is quad residue $\pmod{n}$ $\longleftrightarrow$ a is quad residue $\pmod{p}$ for each prime divisor p of n

i.e $\left(\frac{a}{n}\right) = 1$ does not imply a is quad residue $\pmod{n}$

$\because$ Hensel's Lemma.

$\left(\frac{a}{n}\right) = -1$ does imply a is quad non-residue $\pmod{n}$

e.g $\left(\frac{6}{35}\right) = \left(\frac{6}{5}\right)\left(\frac{6}{7}\right) = \left(\frac{1}{5}\right)\left(\frac{-1}{7}\right) = -1$

therefore 6 is quad non-residue of 35

Jacobi is multiplicative

① $(a,n) = 1$, $(b,n) = 1$ then $\left(\frac{ab}{n}\right) = \left(\frac{a}{n}\right)\left(\frac{b}{n}\right)$

② m, n are odd, $(a, mn) = 1$ then $\left(\frac{a}{mn}\right) = \left(\frac{a}{m}\right)\left(\frac{a}{n}\right)$

$\left(\frac{-1}{n}\right) = (-1)^{\frac{1}{2}(n-1)}$

$\left(\frac{2}{n}\right) = (-1)^{\frac{1}{8}(n^2-1)}$

$\left(\frac{m}{n}\right)\left(\frac{n}{m}\right) = (-1)^{\frac{1}{4}(m-1)(n-1)}$

$\because$ $\frac{1}{2}(n-1) \equiv \frac{1}{2}(n_1-1) + \frac{1}{2}(n_2-1) \pmod{2}$

because $\frac{1}{2}(n_1-1)(n_2-1) \equiv 0 \pmod{2}$

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335 is quad residue $\pmod{2999}$

$$\left(\frac{335}{2999}\right) = -\left(\frac{2999}{335}\right) = -\left(\frac{-16}{335}\right) = -\left(\frac{-1}{335}\right)$$